



2 A school with a large number of students is updating its logo. Each student has designed a new logo and two teachers have each awarded a mark out of 50 for each logo. The marks awarded to a random sample of 12 students are shown in the following table.

| Student   | $A$ | $B$ | $C$ | $D$ | $E$ | $F$ | $G$ | $H$ | $I$ | $J$ | $K$ | $L$ |
|-----------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Teacher 1 | 36  | 38  | 40  | 36  | 22  | 34  | 45  | 44  | 48  | 35  | 28  | 30  |
| Teacher 2 | 38  | 42  | 32  | 41  | 32  | 41  | 42  | 50  | 36  | 44  | 42  | 41  |

One of the students claims that Teacher 2 is awarding higher marks than Teacher 1.

- (a) Carry out a Wilcoxon matched-pairs signed-rank test, at the 5% significance level, to test whether the data supports the claim. [7]

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